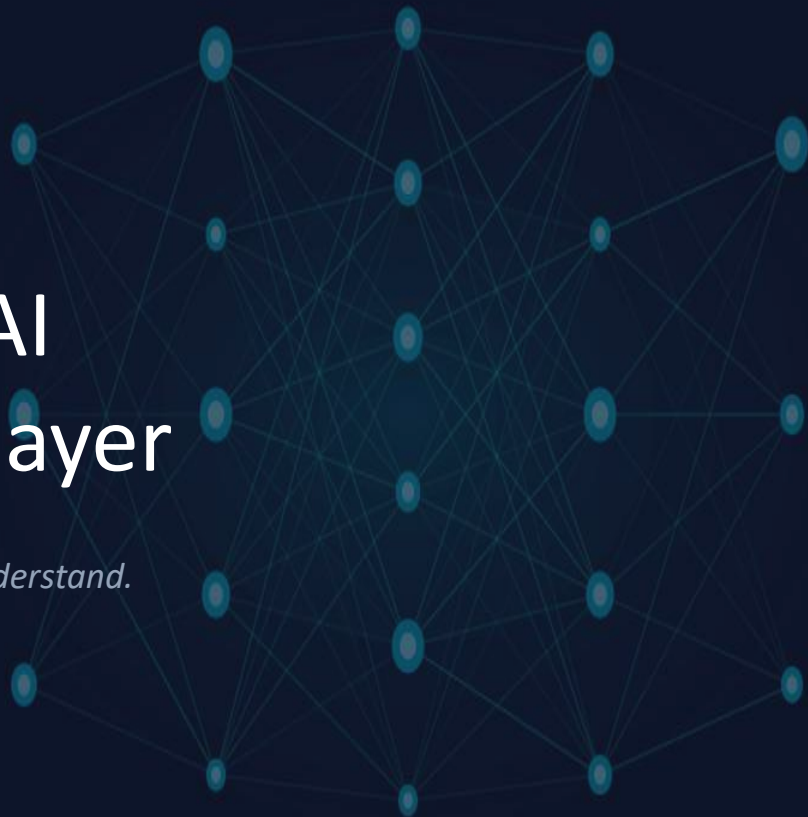


Building a User and AI optimised semantic layer

How to make models that both people and AI can understand.



ONE DAY



ONE DAY



Mihaly Kavasi

Principal Solution Architect

One Day Business Intelligence



- ✓ 10+ Years of Power BI Experience
- ✓ 9+ Year of Training Experience
- ✓ Microsoft Certified Solution Expert – Data Management and Analytics
- ✓ Microsoft Certified Trainer
- ✓ Azure Enterprise Data Analyst
- ✓ Fasttrack Recognized Solution Architect



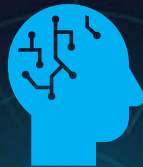
Welcome to the New world of AI everything!

Predictions

- At **Davos** a few days ago **Dario Amadei** (CEO of Anthropic) and **Demis Hassabis** (CEO of Google Deepmind) talked about how imminent is that AI will replace knowledge workers.
- We hear about Amazon in the last few month fired 30.000 employees, and call centres are getting replaced by AI.

Experience

- Recent study shows that 70% of organizations did not see any productivity gain from AI last year!
- Another study shows that 80%+ of AI projects fail or abandoned.



Understand AI a bit!

How we should think about AI “Thinking”!



How Humans think!

Daniel Kahneman described in his book "Thinking, Fast and Slow":

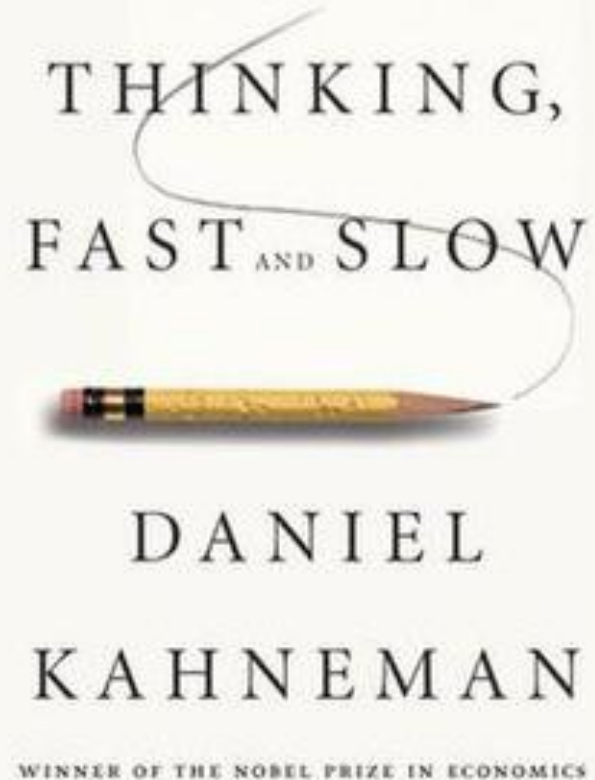
- "**System 1**" that is fast, instinctive and emotional
- "**System 2**" that is slower, more deliberative, and more logical

Walking, driving, shopping, generally going through life on "auto-pilot", that is **System 1**.

But if you are learning something new or solve a non-trivial problem, it is **System 2**. When you are using **System 2** you feel it. It becomes hard and tiring.

Opinion:

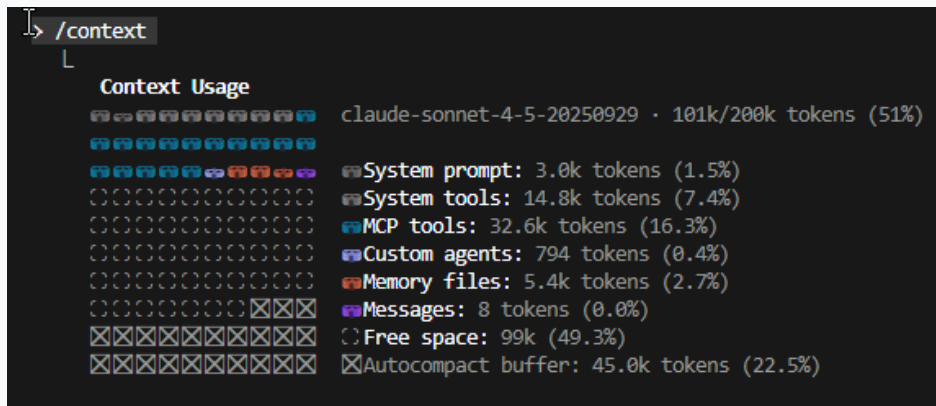
LLMs are not thinking machines, but pattern matching algorithms. The reason we feel like they are thinking is because most of the time our brain is also mainly doing pattern matching. It is struggling with ambiguity and novelty.





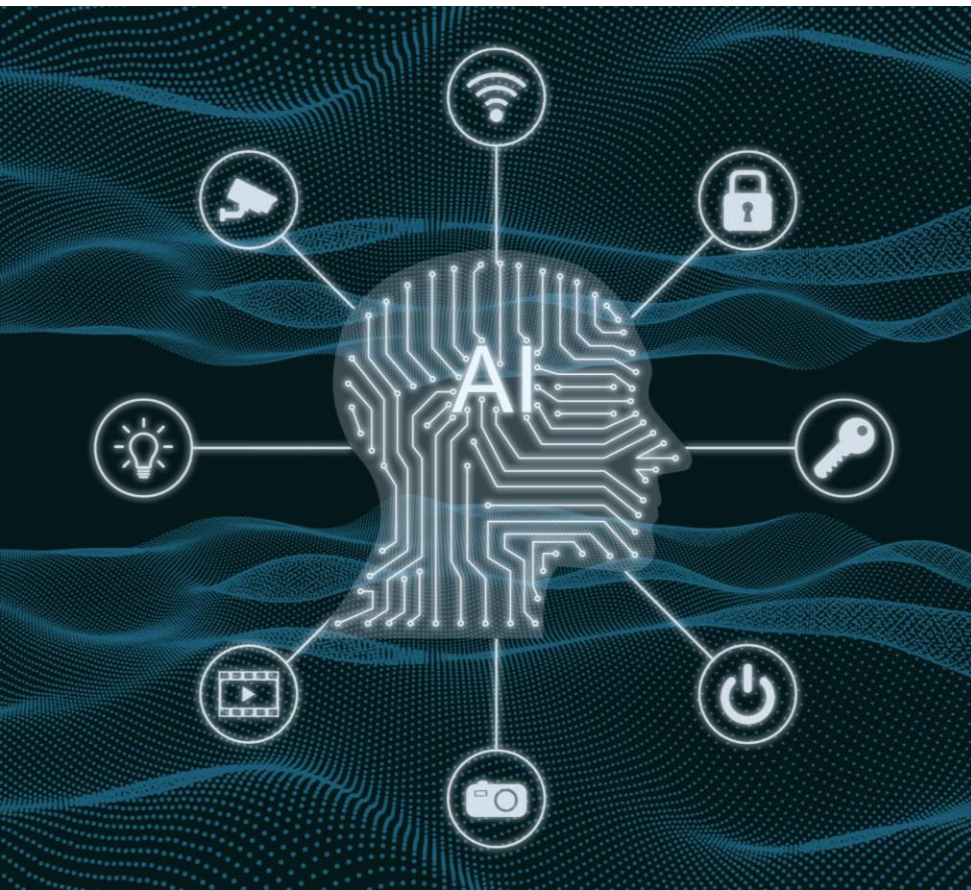
Understanding Memory

- **Context window** is what we call AI-s short term memory.
- Generally, between **200.000 to 1.000.000** token.
- Agents tend to behave erratically when getting closer to the end of their context windows.
- **Auto Compaction** can destroy important context decreasing the quality of their response





What kind of work AI will most likely replace?

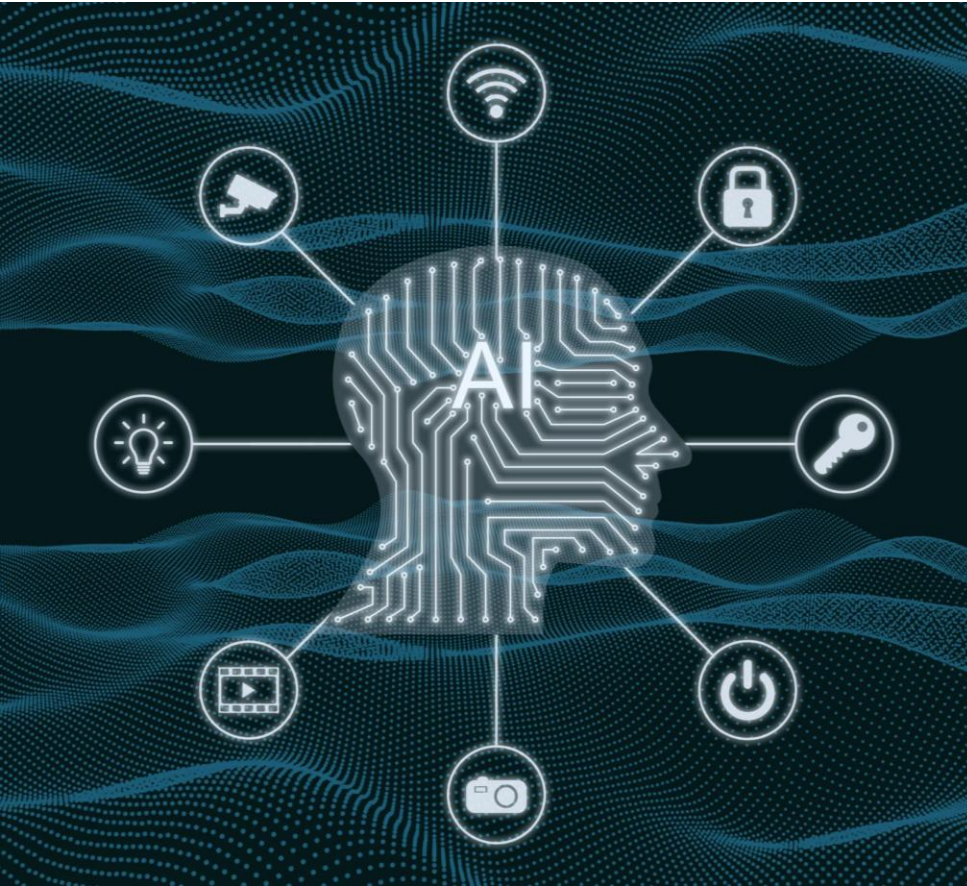


- Jobs and tasks that does not benefit from dealing with uncertainty, subjectivity or novelty.
- Where good enough is ok.
- Even then AI, because it is a **non-deterministic token generator**, need to be supplemented with robust systems, controls and processes.
- If interested come to **SQLBits 2026** where I will talk about these systems together with **Scott Bell**

Event	Session	Date
SQLBits 2026	Accepted Building Context, Not Vibes Practical AI Augmented Data Engineering (Part 2)	📅 22 Apr - 25 Apr 2026
	Accepted Building Context, Not Vibes Practical AI Augmented Data Engineering	📅 22 Apr - 25 Apr 2026



What Does it Mean For a Semantic Model Development?



- **Context is KING!**
- Anytime the AI must assume, or guess opens the possibility for misinterpretation.
- This won't be solved by better models.
- **If a human can easily understand our model so will the AI and vice versa.**



Let's go back in time!

How we got here!



Emergence of Self-Service

Business

- Agility
- Access
- Time to insight
- Ease of use

IT/BI

- Stability
- Security
- Avoid Rework
- Ease of Integration

There are conflicting interests between Business and IT.

This results in the emergence of shadow BI from Business and strict access rules from IT.

Self-Service BI solutions emerged to try to ease the collaboration.



How to split the responsibilities between Business and IT/BI?

Business is responsible for:

- Validating concepts/prototypes
- Creating reports
- Design
- Adoption support

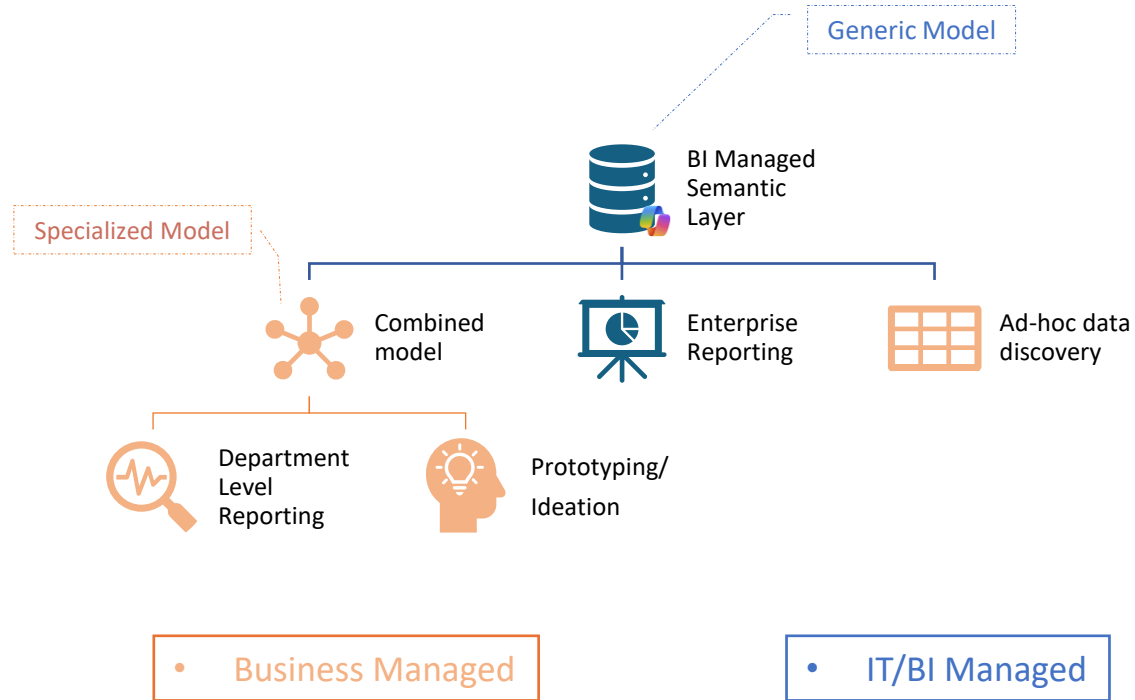
IT/BI is responsible for:

- Data quality
- Data freshness
- Security
- Correctness of insights



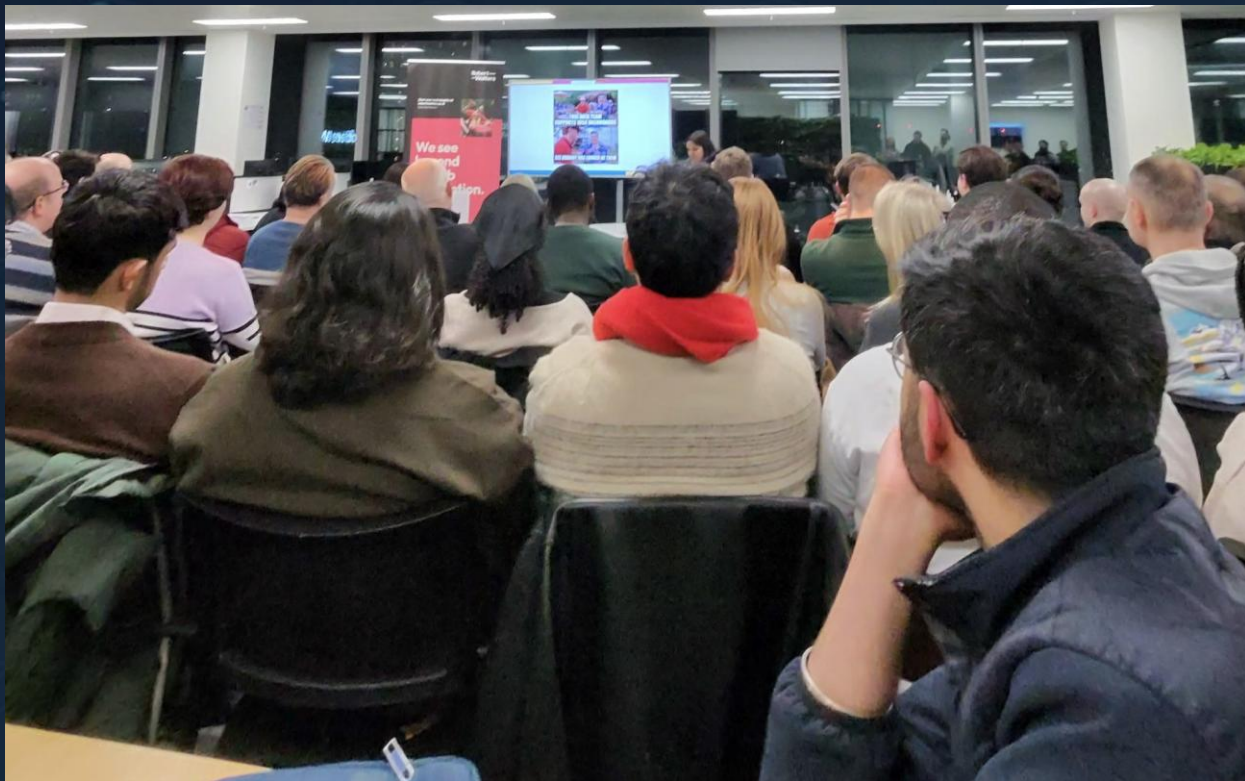
Why should we create a user-friendly semantic layer?

- Semantic layer in Power BI is the **most important part of any analytics solution.**
- Eliminate unnecessary duplications of effort and data.
- **Single source of truth.**
- **One good dataset can be a source of 10s or even 100s of reports**
- **Less maintenance and change management**
- **More time to optimize**





To avoid creating a 1000 reports that nobody looks at!





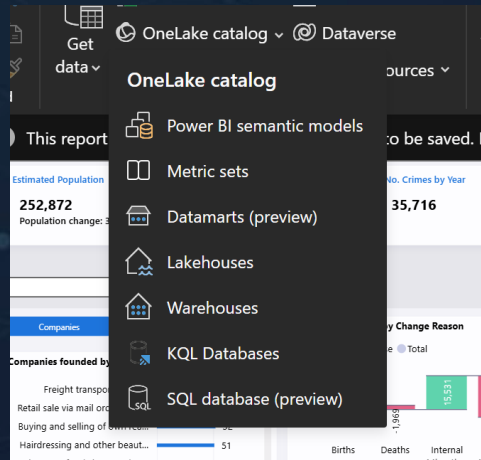
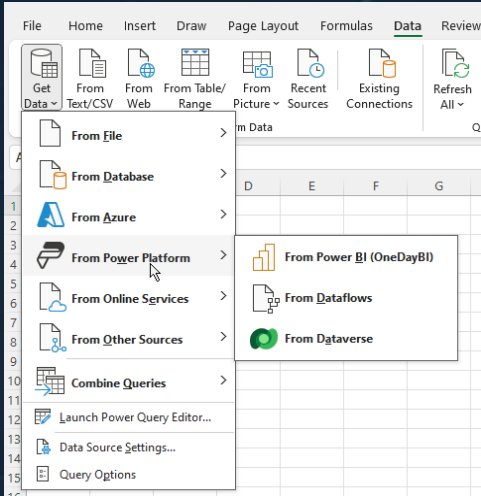
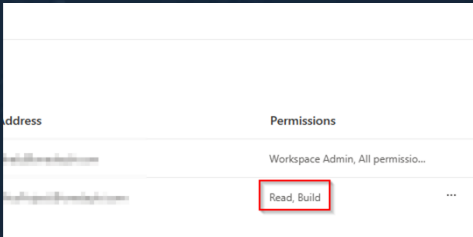
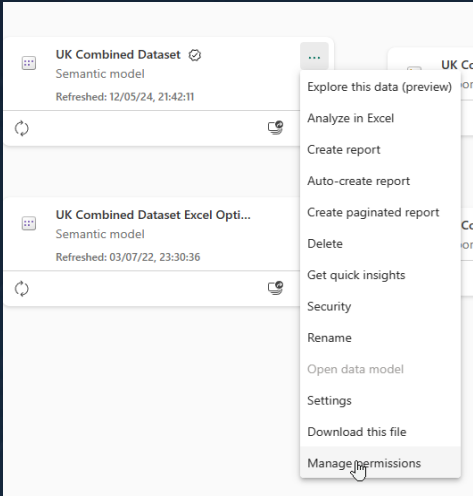
Difference between creating for yourself and others

Building an optimized semantic layer



When you want to create a semantic layer....

Share the dataset but not the workspace to provide access but maintain security



... make sure your audience knows which one is the best for their data needs

- Use the certification system
- Set Discoverability
- Add Semantic Layer Descriptions

Endorsement and discovery

Help coworkers find your quality content by endorsing this semantic model and making it discoverable. [Learn more](#)

☐ None

This semantic model will appear in search results but isn't endorsed.

☒ Promoted

When you're ready to distribute the semantic model to your coworkers, promote it to let them know.

☐ Certified

Certify your semantic model to show coworkers that it's been reviewed and meets your org's certification criteria. [How do I get my semantic model certified?](#)
This Datamart is a core source of organizational data records.

☒ Make discoverable

Allow users without access to this semantic model to discover it and request permissions to access the data. [Learn more](#)

This dataset will be made discoverable. Others in your org will be able to find it by such details as name, tables, columns, etc. [Learn more](#)

Apply

Discard

OneLake data hub

Discover data from across your org and use it to create reports

This dataset combines publicly available data from the UK Population, housing, education, crime, company related information available. Data is refreshed monthly. Row level security is not applied.

		Refreshed	Location	Endorsement	Sensitivity
UK Combined Dataset	Kávási Mihály	12/05/24, 14:38:31	UK Combined Dataset	Promoted	-
Power BI Tenant Settings	Kávási Mihály	04/08/22, 21:26:16	SelfService BI Blog	Certified	-
Power BI data sources	Kávási Mihály	04/08/22, 21:27:42	SelfService BI Blog	Certified	-
UK Combined Dataset Excel Optim...	Kávási Mihály	03/07/22, 23:30:36	UK Combined Dataset	-	-
Company Master Data	Kávási Mihály	28/06/22, 23:23:07	UK Combined Dataset	-	-
Management Demo	Kávási Mihály	29/04/18, 23:42:49	Demo	-	-
Management Demo	Kávási Mihály	17/04/18, 00:30:25	Demo	-	-

Filter by keyword Filter

Connect Cancel

Power BI Datasets

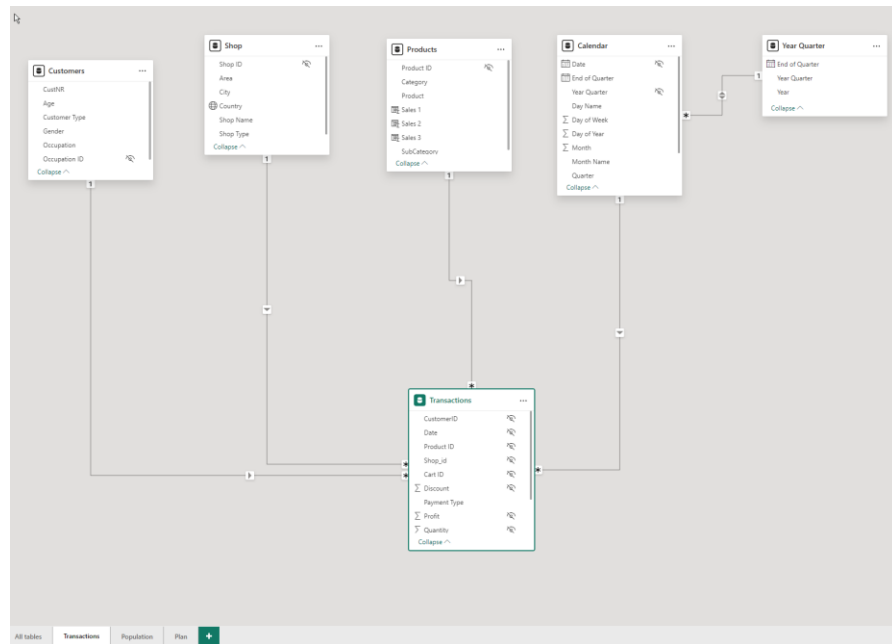
com

Competitor_Analysis Workspace: Demo Owner: Kávási Mihály Refreshed: 17/04/2018, 00:32:05 Tables in this dataset (0) Reports using this dataset (0) + Insert PivotTable + Insert Table Promoted
UK Combined Dataset Workspace: UK Combined Dataset Owner: Kávási Mihály Refreshed: 28/06/2022, 23:39:25 Tables in this dataset (0) Reports using this dataset (0) + Insert PivotTable + Insert Table



Naming Conventions and Standard

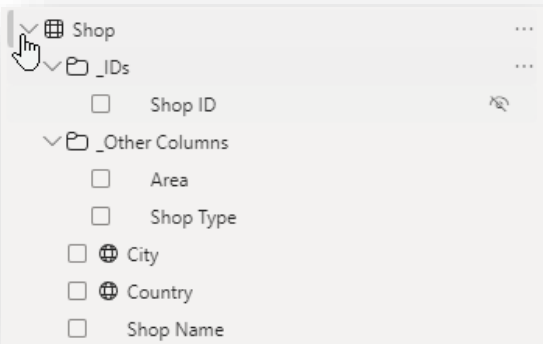
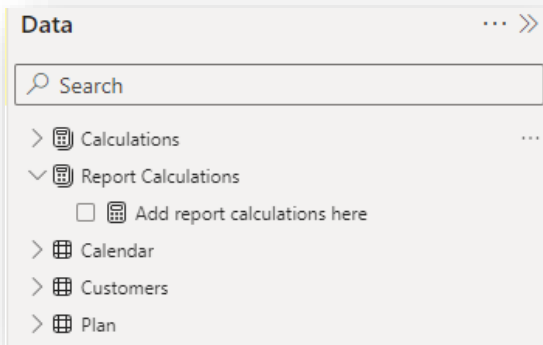
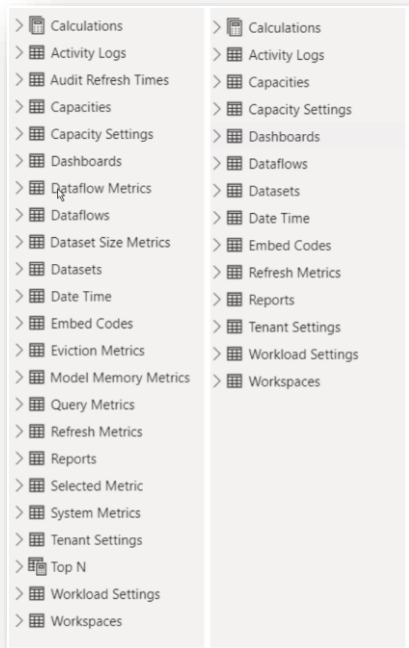
- **Unambiguous and business specific naming standards** for tables, columns, measures, hierarchies and relationships
- **Ensure that there is only one of each calculation.** Do not have 10 slightly different versions of Sales calculation.
- **Make sure that additional filters are not needed to display the right values using the measure.** For example, the need to filter out returned sales in the visual filter to give you the proper sales volume, have a measure that already provides that.
- **Check if measures work with all necessary dimensions,** add that in the description.
- Have detailed descriptions for each measure, table (optionally columns) to explain their meaning, usage and edge cases.
- Remove unnecessary columns
- Create Layouts for each Fact table in the Model view





Useful content visible and organized

- Follow **Analytical Data Modelling** best practices (Star schema)
- **Hide all the Fact tables** (they should only contain facts and id columns)
- **Hide all the id/relationship columns**
- **Do not create measures in data tables**
- **Create a Calculations table** for model metrics
- **Create a Report calculations table** for report metrics
- **Organize your measures** into logical folders
- **Surface your most important dimension attributes** by create folders for the rest



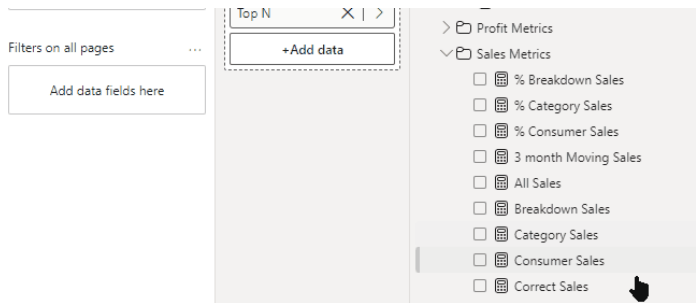


Make your measures understandable

- Give meaningful names
- **Only keep visible the reusable measures**
(hide any intermediary or report specific measures)
- Format your measures
- As a minimum add the measure definition to the description
- Optimally create meaningful descriptions of each measure.
- Semantic Layer measures should not contain design related logic (like +0)
- Do not leave broken or unnecessary measures in the model.

```

1 3 month Moving Sales =
2 IF([Total Sales],
3  CALCULATE (
4      [Total Sales],
5      DATESBETWEEN (
6          'Calendar'[Date],
7          CALCULATE ( MAX ( 'Calendar'[Date] ), DATEADD ( 'Calendar'[Date], -3, MONTH ) ) + 1,
8          MAX ( 'Calendar'[Date] )
9      )
10 )
11 )
    
```



Data should be clean and aligned

- Semantic Layer by default only contain metadata related information
- Nothing about the data contained into model
- You can use Vertipak Analyzer in DAX Studio to analyze your model.
- Make sure you do not have referential integrity violations
- As a next step you can extract the VPAX data and integrate it into your semantic model through Tabular Editor to enrich your model.
- AI will be able to understand it much better.

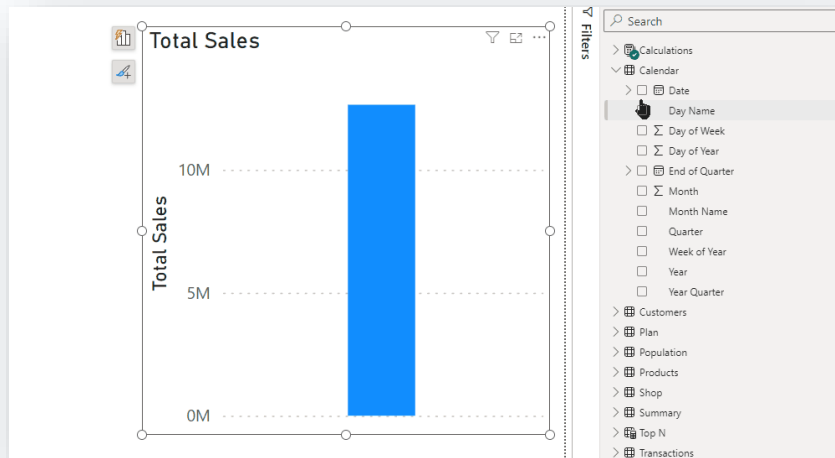
Tables	Columns	Relationships	Partitions	Summary
Name	Cardinality	Table Size	Col Size	Data Dictionary Hier Size Encoding Data Type Ri Violations User
Adventureworks	7,438	953,035	953,035	29,760 813,755 109,520 Many - 1
Adventureworks	9	18,988	18,988	16 18,748 224 Many - -
Adventureworks	133	186,450	186,450	600 182,682 3,168 Many - 17
Adventureworks	133	186,450	186,450	600 182,682 3,168 Many - 1
Adventureworks	133	186,450	186,450	600 182,682 3,168 Many - 3
Adventureworks	1	17,244	17,244	8 17,172 64 Many - -
Adventureworks	6,430	882,975	882,975	56,624 669,871 156,480 Many - 9
Adventureworks	67,936	24,465,431	24,465,431	3,270,256 20,177,535 1,017,640 Many - -
Adventureworks	67,936	10,078,544	10,078,544	3,275,800 5,513,336 1,289,408 Many - 4
Adventureworks	365	357,850	357,850	6,896 339,770 11,184 Many - -
Adventureworks	365	409,630	409,630	6,912 391,230 11,488 Many - -

Tables	Columns	Relationships	Partitions	Summary			
Table / Relationship	Size	Max From Cardinality	Max To Cardinality	1:M Ratio %	Missing Keys	Invalid Rows	Sample Violations
Adventureworks	188,480	45,950	3,683,302	2.97%	18,536	544,253	
Adventureworks	12,872	6,430	6,430	0.97%	0	0	
Adventureworks	9,464	4,726	4,761	0.74%	13	85	
Adventureworks	15,744	3,510	67,936	1.93%	22	62,455	
Adventureworks	7,032	3,510	6,440	0.18%	1	19	GB07PG80700002437
Adventureworks	4,624	1,729	66,046	1.88%	0	0	
Adventureworks	1,992	989	6,430	0.18%	1	1	GB07
Adventureworks	1,472	549	67,936	1.93%	1	1	(blank)
Adventureworks	200	117	3,048	0.09%	1	29,227	0
Adventureworks	200	93	7,640	0.22%	2	11	GB06, GB07
Adventureworks	80	71	133	0.00%	1	204	(blank)
Adventureworks	32	25	120	0.00%	13	3,038	GB04, GB13, GB28
Adventureworks	32	32	38	0.00%	0	0	
Adventureworks	24	16	115	0.00%	1	204	(blank)
Adventureworks	24	20	133	0.00%	1	29,750	(blank)
Adventureworks	16	13	100	0.00%	0	0	
Adventureworks	8	4	8	0.00%	0	0	
Adventureworks	8	1	4	0.00%	0	0	
Adventureworks	9,968	3,346	67,936	3.15%	1	1	

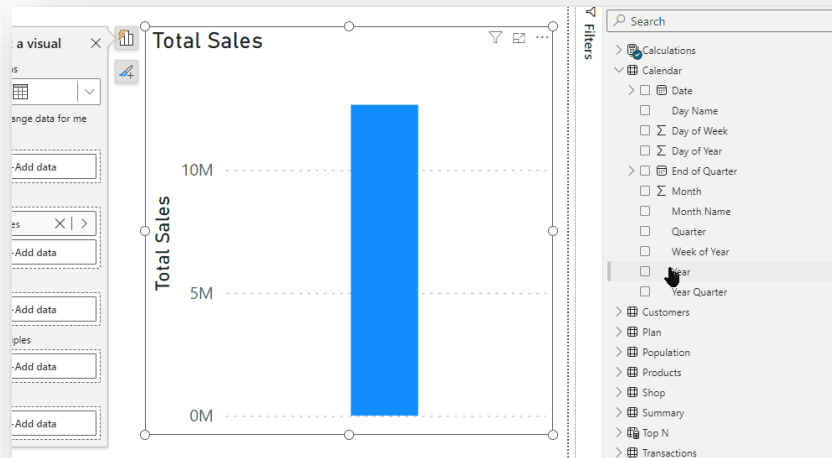


Make sure the data behaves as expected - Sorting

Sorting not set ✗



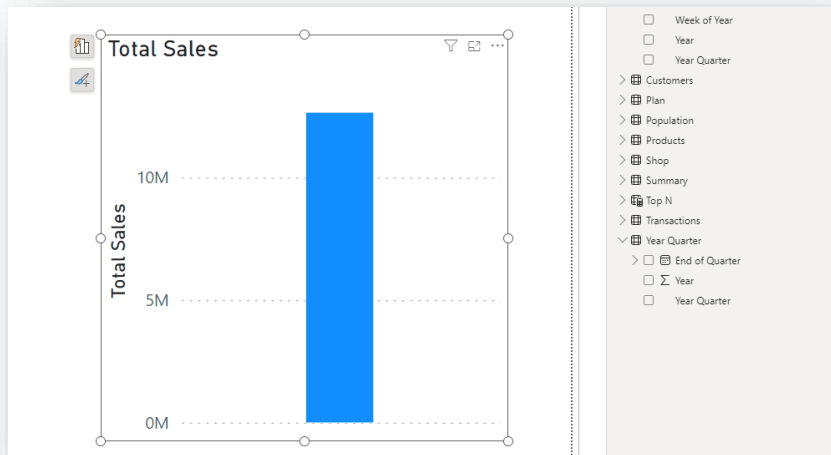
Sorting set ✓



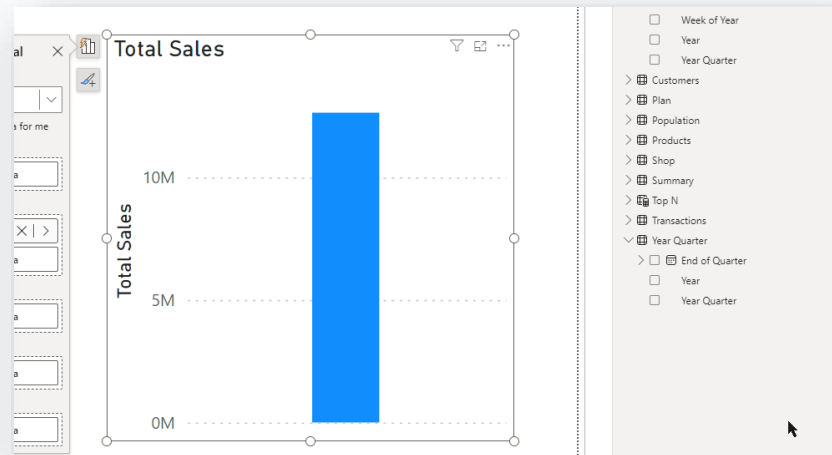


Make sure the data behaves as expected - Summarization

Summarization not set correctly ✗



Summarization set correctly ✓

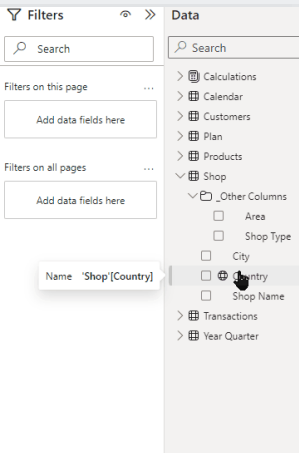




Make sure the data behaves as expected - Data Category

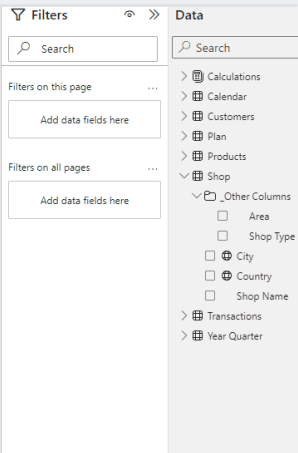
Data category not set correctly ✗

anvas.



Data category set correctly ✓

anvas.

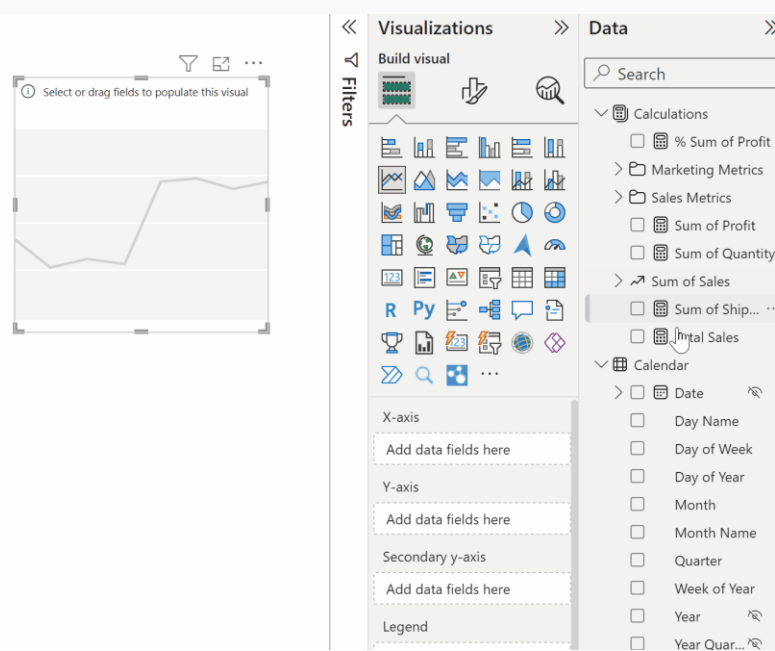
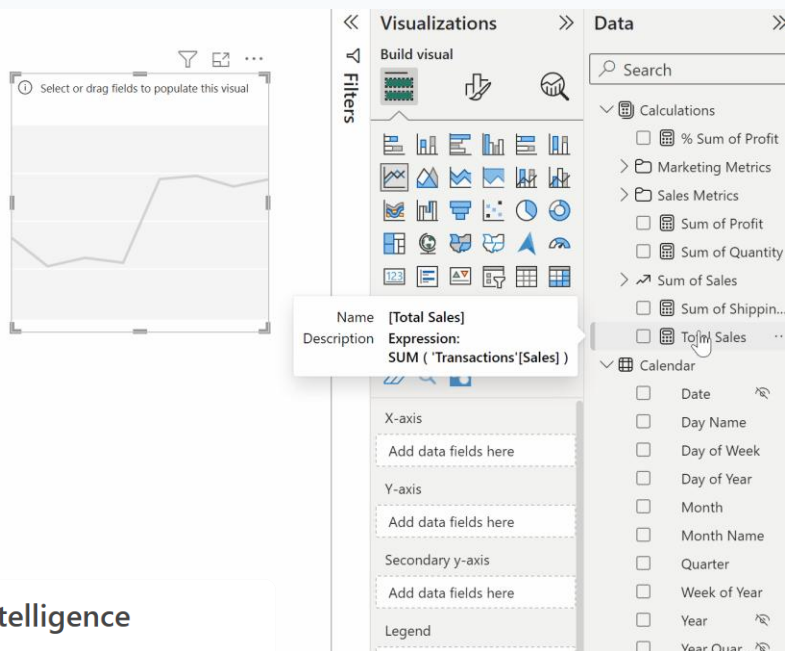




Make sure the data behaves as expected - Auto DateTime

Auto DateTime off ✗

Auto DateTime on ✓



Time intelligence

☒ Auto date/time ⓘ [Learn more](#)

Warning this is controversial suggestion!



Accommodate Excel Features - Show Details

Detail Rows expressions not set ✗

The screenshot shows a DaxStudio window with a table named 'Computer Parts'. The table has two columns: 'Row Labels' and 'Total Sales'. The data is displayed in a standard Excel-like grid. The status bar at the bottom indicates 'Ready' and 'Accessibility: Good to go'.

Row Labels	Total Sales
FIC	64269.573
FOXCONN	53927.517
FUJITSU	1467385.07
HITACHI	57187.095
HP	643651.7761
HTC	34254.855
HYUNDAI	47753.706
IBM	40395.687
MESH COMPUTERS	20340.903
MICROSOFT	36548.412
RAZER	29772.585
SAMSUNG	1325572.643
SAPPHIRE TECHNOLOGY	32481.3
SGI	39990.756
SHUTTLE	40501.623
SIRAGON	25545.933
SUPERMICRO	50914.848
TOSHIBA	658595.5255

Detail Rows expression set ✓

The screenshot shows a DaxStudio window with a table named 'Computer Parts'. The table has two columns: 'Row Labels' and 'Total Sales'. The data is displayed in a standard Excel-like grid. The status bar at the bottom indicates 'Ready' and 'Accessibility: Good to go'.

Row Labels	Total Sales
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HTC	34254.855
HYUNDAI	47753.706
IBM	40395.687
MESH COMPUTERS	20340.903
MICROSOFT	36548.412
RAZER	29772.585
SAMSUNG	1325572.643
SAPPHIRE TECHNOLOGY	32481.3
SGI	39990.756
SHUTTLE	40501.623
SIRAGON	25545.933
SUPERMICRO	50914.848
TOSHIBA	658595.5255



Continuously Optimize Performance

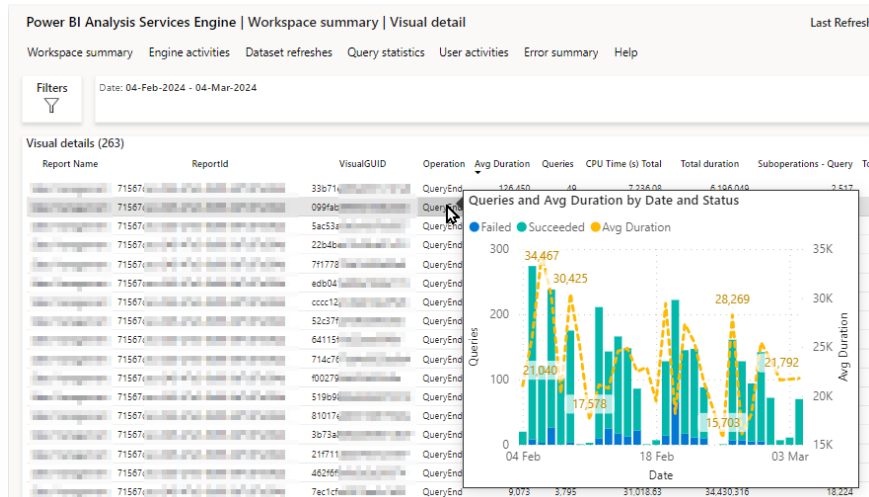
During Development:

- Use tools like Performance Analyzer and DAX Studio to test and optimize the performance.

StartTime	Type	Duration	User	Database	Query
09:59:17	DAX	2ms	DESKTOP...	datamodel	DEFINE VAR _DSOFilterTable = FILTER(KEEPFILTERS(VALUE...
09:59:17	DAX	11ms	DESKTOP...	datamodel	DEFINE VAR _DSOFilterTable = FILTER(KEEPFILTERS(VALUE...
09:59:17	DAX	6ms	DESKTOP...	datamodel	DEFINE VAR _DSOFilterTable = FILTER(KEEPFILTERS(VALUE...
09:59:16	DAX	10ms	DESKTOP...	datamodel	DEFINE VAR _DSOFilterTable = FILTER(KEEPFILTERS(VALUE...
09:59:16	DAX	5ms	DESKTOP...	datamodel	DEFINE VAR _DSOFilterTable = FILTER(KEEPFILTERS(VALUE...
09:59:16	DAX	2ms	DESKTOP...	datamodel	DEFINE VAR _DSOFilterTable = FILTER(KEEPFILTERS(VALUE...
09:59:14	DAX	2ms	DESKTOP...	datamodel	DEFINE VAR _DSOCore = VALUES(Top N(Top NJ) VAR _DSOBodyLimited = SAMPLE(1002, _DSOCore, 'Top N(Top NJ, 1) EVAL...
09:59:03	DAX	1ms	DESKTOP...	datamodel	DEFINE VAR _DSOCore = DISTINCT(Top N(Top NJ) VAR _DSOPrimaryWindowed = TOPN(501, _DSOCore, 'Top N(Top NJ, 1) E...

During Production:

- Use Log Analytics to continuously monitor performance and proactively optimize based on utilization patterns



You can find me:

- [Website: www.onedaybi.com](http://www.onedaybi.com)
- [Blog: www.selfservicebi.co.uk](http://www.selfservicebi.co.uk)
- [LinkedIN: https://www.linkedin.com/in/kavasimihaly/](https://www.linkedin.com/in/kavasimihaly/)
- [GitHub: https://github.com/KavasiMihaly/](https://github.com/KavasiMihaly/)
- [My Tabular Editor Scripts: https://github.com/KavasiMihaly/SelfServiceBIBlog/tree/main/TabularEditor](https://github.com/KavasiMihaly/SelfServiceBIBlog/tree/main/TabularEditor)